



Estimating earthquake early warning effectiveness via blind zone sizes: a case study of the new seismic network in Chinese mainland

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Abstract

The China Earthquake Administration (CEA) has launched an ambitious nationwide earthquake early warning (EEW) system project, now fully operational as of July 2024. This system, the largest EEW network in the world, comprises approximately 15,000 seismic stations. In approximately 50%, 30% and 20% of Chinese mainland, the inter-station distance will soon be smaller than 50 km, 25 km and 15 km, respectively. The expected effectiveness of this EEW system can be quantified via the metric determined from the radius of the blind zone, which refers to the area near the epicenter where there is insufficient time to issue a warning before the arrival of strong *S*- and surface waves. This study uses a theoretical network-based method together with Monte Carlo simulation to obtain the spatial distribution of the blind zone radii and their associated uncertainties for the new seismic network based on its configuration. We find that the densified new seismic network is expected to have excellent EEW performance as the area covered by small blind zones with a radius ≤ 30 km increases dramatically from approximately 2–22%, that is by 2.4 million km² inside Chinese mainland. This improvement reduces regions affected by large blind zones and translates into substantial gains in lead time (available warning time). Compared to the previous network layout, lead times are expected to increase by approximately 20 s, 10 s, and 5 s for approximately 50%, 30%, and 10% inside Chinese mainland, respectively. We also conducted a preliminary cost-outcome evaluation of the new EEW system from the perspective of blind zones. Continuing to increase the density of stations in some key regions with blind zone radii ranging from 15 to 40 km is still necessary to control the unexpected expansion of blind zones due to possible (and common) stations failure. Our work provides insights into the expected performance of the upcoming EEW network in Chinese mainland, and our proposed evaluation approach is broadly applicable for predicting the performance of EEW systems during their planning, design, and implementation stages.

Keywords Earthquake early warning (EEW) · Blind zone · System alert trigger latency · Lead time · Seismic network assessment · Chinese mainland

1 Introduction

The blind zone represents the geographic area around the earthquake epicenter where there is insufficient time for issuing warnings before the onset of intense *S*- and surface seismic waves (Kuyuk and Allen 2013). The concept of the blind zone is typically linked to the timing of the initial alert triggered by an Earthquake Early Warning (EEW) system. The blind zone, also known as the no-warning zone, no-alert zone or late-alert zone, directly reflects the limits of an EEW system's ability to provide timely alerts (with positive lead time or available warning time) to regions at risk. This is a crucial aspect of an effective EEW system, enabling the implementation of risk mitigation measures to reduce potential damage and save lives (Picozzi et al. 2015; Pan et al. 2019; Wu et al. 2019; Cremen et al. 2022; Zhang et al. 2021, 2023). A smaller blind zone denotes a more efficient and robust EEW system, ensuring that vulnerable regions receive advance warnings of impending destructive seismic waves. However, even the blind zone itself, regardless of its size, may contain regions at high risk. The densification of a seismic network reduces blind zones of an EEW system (Kuyuk and Allen 2013).

The concept of the blind zone is generally associated with the timing of the first triggered alert by an EEW system. Although the initial alert may contain significant estimation errors in various parameters, the accuracy of the information improves as additional station data becomes available, allowing for more precise updates in subsequent alerts. However, the recipients of the alert typically initiate emergency actions upon receiving the first alert. The system alert trigger latency defined as the time T_{tri} between the earthquake's origin and the system's first alert, is considered the primary factor contributing to the existence of the blind zone for shallow earthquakes. T_{tri} is determined by many factors, e.g., network density, focal depth, seismic velocity structure of the crust, speed of data transmission and system processing (Kuyuk and Allen 2013; Yang et al. 2015; Picozzi et al. 2015; Guo et al. 2016; Li and Wu 2016; Festa et al. 2018).

The main factors ensuring the feasibility of an EEW system include lead time, expected damage, adopted EEW algorithms, and others. Among these, the blind zone radius plays a crucial role by determining the length of the lead time and the size of vulnerable regions that cannot be warned in advance. Therefore, the blind zone radius r is the simplest and one of the most effective proxies of the overall performance of an EEW system. The radius of the blind zone plays a crucial role in assessing the effectiveness of EEW systems. Although it is impossible to act in a timely or effective manner within the blind zone, an EEW system is designed to be useful for the sites outside the blind zone, particularly those near the fault rupture that may still experience severe seismic damage. This is especially relevant for earthquakes with hundred kilometers ruptures (Böse et al. 2018, 2021; Ma et al. 2012; Li et al. 2018, 2020). It is geometrically unavoidable for shallow earthquakes to have blind zones with radii ranging from 10 km to 20 km, regardless of network density. This limitation arises from the inherent constraints of data transmission speed and system processing time. For instance, most EEW algorithms require at least 3 s of *P*-wave data to perform the necessary calculations, followed by an additional 1 to 3 s for processing and data transmission. During this period, *S*-waves and surface waves, typically traveling at velocities around 3.5 km/s, continue to propagate and cause damage before an alert can be issued (Kuyuk and Allen 2013; Wald 2020). As a result, optimizing an EEW system primarily focuses on minimizing the size of the blind zone to expand the area covered by early warnings. In this study, we

focus on reducing the blind zone size by improving the network geometry, with the aim of evaluating the total area impacted by the reduction in blind zones.

Inspired by the developments of the EEW systems, and also the benefit from them, in other regions worldwide (Allen et al. 2009; Satriano et al. 2011a; Strauss and Allen 2016; Allen and Melgar 2019; Allen and Stogaitis 2022), Chinese mainland has been continuously developing and evolving its EEW systems at the city/infrastructure and regional/provincial scales in order to improve the capability for earthquake risk mitigation in the past three decades (e.g., Li et al. 2004; Peng et al. 2011; Li 2014; Zhang et al. 2016; Peng et al. 2020, 2021, 2022). To further extend the scale and applications of the previous seismic network, the National System for Fast Report of Intensities and Earthquake Early Warning Project of China launched by the China Earthquake Administration (CEA) has started a nationwide network construction project with a total investment of approximately 1.87 billion RMB from 2018 (290 million USD; see Data and Resources). Through this project, the number of seismic stations will increase from approximately 2,000 to 15,000, with data transmission for all stations planned to have very little latency (Jiang and Liu 2016). Considering the specific characteristics of seismic hazards and urban distribution in China, the new EEW system will adopt a hybrid model that integrates both onsite and regional warning approaches (Jiang and Liu 2016).

With the continuous installation of new stations and the gradual upgrading of previous ones, the EEW system has undergone trial operations for several years in regions such as Beijing, Shanghai, Xinjiang, Sichuan, Yunnan, and Fujian (Peng et al. 2020, 2021, 2022). For example, in Xinjiang, it is reported that approximately 50% of the region has a T_{tri} of less than 10 s (Yan et al. 2023). For three notable earthquakes in the region—the 2021 Baicheng M_S 5.4, 2022 Changji M_S 4.8, and 2024 Wushi M_S 7.1 earthquakes—the reported T_{tri} values were approximately 4 s, 8 s, and 12 s, respectively. Notably, the 2024 Wushi M_S 7.1 earthquake involved a regional warning, as it occurred 22 km from the Xinjiang seismic network (Zhao et al. 2024; Yan and Chen 2024; Yan and Li 2021). In the Sichuan and Yunnan regions, reports indicate that T_{tri} typically ranges from 5 to 8 s (Li et al. 2024b; Guo et al. 2024). The Sichuan EEW system successfully issued a regional warning for the 2021 Qinghai Maduo M_S 7.4 earthquake, which occurred 55 km outside the Sichuan seismic network, with a T_{tri} of 12 s (Li et al. 2021b). In Hebei and Shanghai, reported T_{tri} values range from 4 to 10 s (Wang et al. 2023; Shao et al. 2024). Overall, during trial operations, the T_{tri} for the new EEW system was between 4 and 10 s for onsite warnings (corresponding to a blind zone radius of approximately 15 to 35 km), while for regional warnings it slightly exceeded 10 s, depending on the epicentral distance from the network edge. The source parameter errors of the first triggered alert were also reported, with location errors around 5 km, magnitude errors around 1 unit, and depth errors around 3 km (Zhao et al. 2024; Yan and Chen 2024; Shao et al. 2024; Li et al. 2024b). In July 2024, the entire project was completed and officially accepted, marking the commencement of full-scale operation of the warning system (see Data and Resources).

The theoretical network-based approach proposed by Kuyuk and Allen (2013) is well-suited for predicting the overall performance of a new EEW system. Unlike approaches focused on single earthquake case analyses within the new network or evaluations of other EEW seismic networks, this method provides a comprehensive assessment of the system's capabilities (e.g., Satriano et al. 2011b; Pazos et al. 2015; Li et al. 2018; Hsu et al. 2018; Peng et al. 2021, 2022). This approach relies exclusively on the spatial distribution of sta-

tions, making it ideal for large-scale system performance assessments. Following a similar approach, Pan et al. (2019) performed a preliminary analysis of system alert trigger latency T_{tri} for the prototype and future EEW systems in Gansu province, China, according to their configurations. Their projections indicated significant improvements in the operational efficiency of the upcoming EEW system in the southeastern region of Gansu. The anticipated system alert trigger latency ranged between 7 and 12 s, corresponding to a blind zone radius spanning from 20 to 40 km. By estimating the latency of the newly established EEW system, it was found that, in the Fujian region, 95% of the area has a T_{tri} ranging between 4 and 6 s, corresponding to a blind zone radius of approximately 15 to 20 km (You et al. 2023).

The theoretical network-based method requires several hypotheses (Kuyuk and Allen 2013; Li and Wu 2016). The minimum number of trigger stations is a critical one among them. The majority of the EEW systems worldwide adopt two to four stations as the minimum number to trigger (Kuyuk and Allen 2013; Wald 2020). However, in operational systems, factors such as station failure near the epicenter (Bormann 2002) or underestimation of ground-shaking hazards (Ma et al. 2012) often reduce the number of functional stations near the epicenter. This, in turn, necessitates a larger number of stations to trigger and expands the blind zone. To address this, Li and Wu (2016) introduced the concept of a “soft” blind zone that surrounds the original (“hard”) blind zone. The “hard” blind zone refers to the blind zone calculated under the assumption that all stations are functioning normally. In contrast, the “soft” blind zone accounts for the possibility of station failures near the epicenter (e.g., stations failing to operate or producing magnitude estimates below the danger threshold). Li and Wu argued that the “soft” blind zone could be effectively mitigated by simply densifying the seismic network. In this context, the “soft” blind zone highlights the practical challenges of ensuring reliable station performance in EEW systems, particularly in regions close to the epicenter. Although an unexpected expansion of the blind zone (“soft” blind zone) is common and challenges the success of an EEW system, so far there is no study that quantitatively investigates how the EEW effectiveness is affected by the “soft” blind zone. Another important issue with the network-based method is that the estimated blind zone radii are subject to parameter uncertainty, which has largely been ignored in the existing applications.

In this study, we predict the EEW performance estimated by the blind zone radii for the new seismic network and assess their uncertainties in terms of probability distributions by employing the Monte Carlo simulation method. We begin with a brief introduction to the new EEW system in Chinese mainland. Then, we predict the improvement of the new network compared with the previous network from the perspective of the spatial distribution of blind zones and their uncertainties. Additionally, we demonstrate how these improvements lead to an increase in lead time (available warning time). Finally, the significance of controlling the “soft” blind zones is discussed quantitatively. Our work offers insights into the anticipated performance of the forthcoming EEW network in Chinese mainland. Furthermore, the proposed evaluation approach can be applied beyond our specific case, for instance for predicting the performance of any EEW system during its planning, design and implementation stages.

2 Data

From January 1, 1970, to June 26, 2022, approximately 1,586,600 earthquakes were observed in China and its adjacent regions, and 91,891 events of them have $M_L \geq 3.0$ as shown in Figure S1. The frequency-depth distribution of these earthquakes reveals three prominent peaks, located at depths of approximately 5 to 10 km, 135 to 140 km, and around 540 km (Figure S1). Notably, earthquakes occurring at depths greater than 400 km are concentrated in Northeast China, bordering Russia and North Korea.

The new seismic network in Chinese mainland consists of a total of around 15,200 stations (Figure S2b): approximately 2,000 datum stations equipped with broadband seismometers and accelerometers with noise levels ranging from -180 dB to -130 dB, 3,200 basic stations equipped with only accelerometers with noise levels around -130 dB and 10,000 ordinary stations equipped with low-cost micro-electro-mechanical system (MEMS) intensity sensors with noise levels ranging from -110 dB to -60 dB (Peng et al. 2022; Li et al. 2024b). For the purpose of this study, we define the inside of Chinese mainland as the region within 100 km outside of the boundaries of Chinese mainland and Hainan Island (approximately 11.5 million km^2 ; Figure S1). The inter-station distance at a given site (assuming a grid of $0.1^\circ \times 0.1^\circ$ resolution) is defined as the average distance from this site to its three closest stations (Kuyuk and Allen 2013). The selection of the three closest stations in this study corresponds to the minimum number of triggered stations required for the EEW process, a topic that will be elaborated on in a later section. In approximately 80% of the area inside Chinese mainland, the inter-station distance of the densified new network is less than 100 km, 50% less than 50 km, and 30% less than 25 km (Figure S3b). The new network will be the main backbone of the largest EEW system in the world. In July 2024, the new EEW system was fully completed and officially approved, signifying the beginning of the system's full-scale operational deployment (see Data and Resources). Moving forward, EEW services will be provided in five key regions (North China, the central China north-south seismic belt, the southeast coast of China, the middle section of the Tianshan Mountains in Xinjiang, and Lhasa in Tibet; Figure S1). In addition to the EEW, the new network will also be utilized for rapid intensity reporting. The seismic intensity will be reported rapidly within minutes from the origin time for the entire Chinese mainland, and the monitoring capability in Chinese mainland will be significantly improved (Li et al. 2024a).

To compare with the new EEW system, we envisage a virtual EEW system by combining the previous high-gain broadband seismic and strong motion networks with a total of approximately 1,000 and 1,100 stations, respectively (as of 2018; Figure S2a). In Chinese mainland, the previous high-gain broadband seismic networks are managed and operated by the China Earthquake Networks Center (CENC) of the CEA and various provincial earthquake agencies, while the previous strong motion networks are managed and operated by the China Strong Motion Networks Center (CSMNC) under the Institute of Engineering Mechanics of the CEA. It is important to note that, among these stations, only high-gain broadband seismic stations have real-time data transmission capabilities; strong motion stations lack this feature. It is worth emphasizing that neither high-gain broadband seismic stations nor strong motion stations were originally designed for EEW purposes. The virtual EEW system is formulated within the previous seismic network under the assumption that all stations possess real-time data transmission capabilities. Only in 55% of inside Chinese

mainland, the inter-station distances of the previous network are less than 100 km, 24% are less than 50 km, and 7% are less than 25 km (Figure S3a).

During the implementation of China's National System for Fast Report of Intensities and Earthquake Early Warning Project, most stations in the previous high-gain broadband seismic and strong-motion networks were upgraded and renovated to incorporate real-time data transmission capabilities, enabling their integration into the new seismic network. Compared with the previous network, the area with inter-station distances smaller than 15 km that is covered by the new network expands by a factor of approximately 10 to reach 17.3% from 1.7% (Figure S3). The areas with inter-station distances smaller than 100 km, 50 km and 25 km will be expanded by factors of 1.4, 2.2 and 4.4 in the near future, respectively. The density of the new network is thus significantly improved.

3 Earthquake early warning performance estimation

In this section, we first introduce the definition of the blind zone radius. Then, to investigate the impact of various factors on the blind zone size, denoted as r , we perform a one-at-a-time sensitivity analysis of r . We also present our proposed method for estimating r using Monte Carlo random sampling. The results include maps showing the mean and the standard deviation of the estimated blind zone radii for the previous and new networks. Additionally, we provide a map illustrating the increase in lead time (available warning time) for the new network compared to the previous network.

3.1 Blind zone radius

Considering the relatively low observational quality associated with the most ordinary stations equipped with MEMS sensors with noise levels ranging from -110 dB to -60 dB, we choose conservatively for this study a minimum number of three trigger stations. In other words, we assume that the EEW system must receive and process signals from at least three triggered stations before it is activated and issues the first alert. Hypothesizing a homogeneous Earth crust model, the P -wave travel time t_p at the third closest station with epicenter distance Δ from a point source with depth H is given by:

$$t_p = \frac{\sqrt{\Delta^2 + H^2}}{V_p} \quad (1)$$

where V_p is the P -wave velocity. Then, with the system latency T for processing and data transmission, the horizontal distance r of S -wave propagation is.

$$r = \sqrt{(t_p + T)^2 V_s^2 - H^2} \quad (2)$$

where V_s is the S -wave velocity. This means that the S -wave has already arrived before the alarm is released at distances smaller than r from the earthquake epicenter. The disk with radius r is called the blind zone in the EEW system.

Equation (2) indicates that the size of the blind zone is determined by the system alert trigger latency T_{tri} , as well as V_S and H . T_{tri} is comprised of t_p and T , with Δ , H and V_P determining t_p . To investigate the impact of five factors (Δ , H , V_P , V_S , and T) on the blind zone size r , we perform a one-at-a-time sensitivity analysis of r by sequentially varying each parameter while maintaining the other four fixed at constant values of $\Delta=30$ km, $H=5$ km, $V_P=6$ km/s, $V_S=3.5$ km/s, and $T=4$ s. These parameters are used exclusively in this section to examine the independent impact of each parameter on r (Fig. 1), illustrating the magnitude and trends of change, including increases, decreases, and saturation. The selection

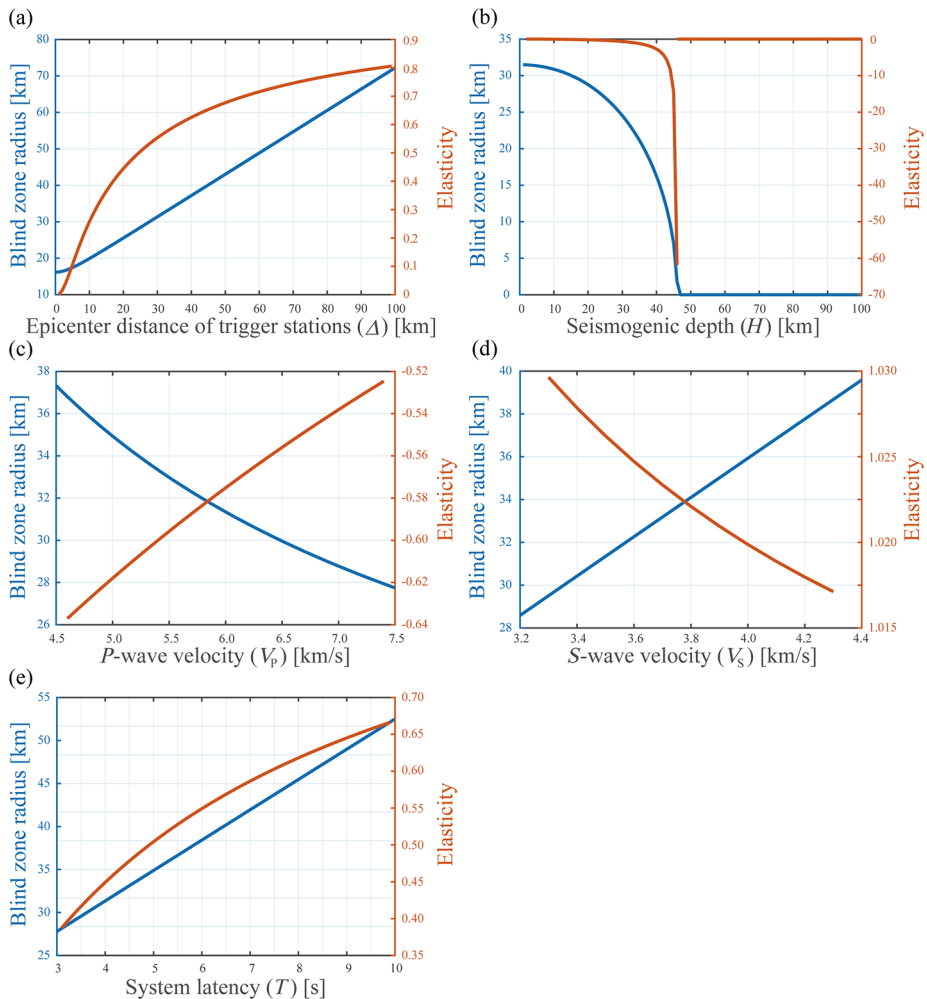


Fig. 1 Dependence of blind zone radius (left axis) on various parameters: (a) epicenter distance (Δ) of trigger stations, (b) seismogenic depth (H), (c) P -wave velocity (V_P), (d) S -wave velocity (V_S), and (e) system latency (T). The blind zone radius has been calculated by sequentially varying each parameter while keeping the remaining four parameters fixed at constant values of $\Delta=30$ km, $H=5$ km, $V_P=6$ km/s, $V_S=3.5$ km/s, and $T=4$ s. In each panel, the red curve (right axis) represents the elasticity of the blue curve (left axis), defined as the relative change in the blind zone radius with respect to the relative change in the variable along the x -axis. For more details, please refer to the main text

of these values is empirical, as they are typical values observed in practice. In subsequent experiments and analyses, we will discuss the selection of these parameters in greater detail, as described in the next section. Figure 1 shows that r has an inverse correlation only with H and V_p . The source depth H contributes significantly to the appearance of a zero blind zone, which occurs when $T_{in}^2 V_S^2 - H^2 \leq 0$.

The impact of each parameter on r was quantitatively analyzed by defining the elasticity (E) of the blind zone radius with respect to each parameter. The elasticity measures the relative change in r with respect to the relative change in a given parameter (Fig. 1), as defined according to:

$$E = \frac{d(\ln r)}{d(\ln x)} \quad (3)$$

where x represents the parameter of interest (e.g., Δ , H , V_S , V_p , and T). The elasticity with respect to Δ , $E(\Delta)$, ranges between 0 and 0.9, indicating that r increases with Δ , but the response is sublinear. A 1% increase in Δ leads to a maximum increase of 0.9% in r , highlighting a moderate sensitivity. For H , the elasticity $E(H)$ behaves nonlinearly. It ranges between 0 and -1 for depths up to 30 km, meaning r decreases nearly proportionally as H increases. Beyond 30 km, $E(H)$ drops abruptly to approximately -60 , reflecting a rapid collapse of r with increasing depth. Around 45 km, $E(H)$ stabilizes at 0, indicating that further increases in H no longer affect r . The elasticity with respect to V_p , $E(V_p)$, remains consistently between -0.64 and -0.52 , suggesting that r decreases as V_p increases, albeit less than proportionally. A 1% increase in V_p reduces r by 0.52–0.64%. In contrast, r exhibits an approximate linear relationship with V_S . The elasticity with respect to V_S , $E(V_S)$, ranges between 1.015 and 1.030, meaning that r increases slightly faster than proportionally with V_S . Specifically, a 1% increase in V_S leads to an approximate 1.02% increase in r . Lastly, the elasticity of r with respect to T , $E(T)$, lies between 0.35 and 0.70, indicating a sublinear positive relationship. A 1% increase in T results in a 0.35–0.70% increase in r , showing moderate sensitivity to latency. These results collectively reveal varying degrees of influence of each parameter on r , with a notable nonlinear behavior as a function of H and an approximate proportional relationship as a function of V_S .

When evaluating the practical influence of various parameters on r , both their elasticity and their range of variations in operational systems must be considered. Empirical observations suggest the range of variations for Δ is approximately 10^1 km (e.g., Figure S3), while it is about 10^0 to 10^1 km for H (e.g., Figure S1), 10^{-1} km/s for V_p (e.g., V_p within a depth of 60 km shown by Han et al. (2022)), 10^{-1} km/s for V_S (e.g., V_S within a depth of 60 km shown by Han et al. (2022)), and 10^{-1} to 10^0 s for T (e.g., Yang et al. 2015). Among these, Δ and H have the greatest impact on r . Although the elasticity of the blind zone radius with respect to Δ is moderate (0 to 0.9), its substantial variability (10^1 km) results in changes for r on the scale of tens to hundreds of kilometers. Similarly, the blind zone radius exhibits extreme elasticity with respect to H , particularly beyond 30 km, where r collapses sharply with elasticities as low as -60 . Variations in H of 10^1 km can reduce r by up to 100 km, underscoring its dominant role in blind zone reduction. In contrast, the blind zone radius shows moderate elasticities with respect to V_p and V_S (-0.64 to -0.52 and 1.015 to 1.030, respectively), but their smaller practical variability (10^{-1} km/s) limits their influence on r with changes of at most 1 to 10 km. Similarly, the elasticity of the blind zone radius with respect to T , with

values between 0.35 and 0.70 and a range of variability for T from 10^{-1} to 10^0 s, contribute changes of the blind zone radius on the order of 5 to 10 km. This combined analysis demonstrates that Δ and H are the most critical factors affecting the blind zone radius, as their variability leads to changes on a far greater scale than other parameters. Therefore, prioritizing improvements in the accuracy and precision of Δ and H in operational systems is essential to effectively minimize blind zone sizes, as even small errors in these parameters can substantially impact the performance of EEW systems.

3.2 Estimating r using Monte Carlo random sampling

It is important to emphasize that Eqs. (1) and (2) represent a significant simplification of the process by which blind zones form in an EEW system. For instance, wave velocities are not uniform and may vary across different paths, and the system alert trigger latency can differ depending on the type of seismic sensors involved. Instead of providing a single-point estimate of the blind zone radius (e.g., Kuyuk and Allen 2013; Li and Wu 2016), we utilize Monte Carlo random sampling to derive the probability density function of the blind zone radius, obtaining a more comprehensive understanding of its variability and uncertainties. Monte Carlo random sampling is a statistical technique used to generate random samples from a probability distribution (Fishman 1996; Kalos and Whitlock 2008; Rubinstein and Kroese 2017). In this study, we employ this approach as a novelty to estimate r for each $0.1^\circ \times 0.1^\circ$ cell of spatial resolution. Firstly, we generate 50,000 random samples for each of the four parameters (H , V_p , V_s , and T) based on given probability density functions (PDFs) and assume a minimum number of three trigger stations to determine Δ . Then, we use these samples along with Eqs. (1) and (2) to obtain 50,000 r estimates. For each grid, we derive statistical parameters such as the mean and standard deviation based on these estimates.

The observed seismicity, depicted in Figure S1 for events with $M_L \geq 3.0$, is utilized to construct the frequency-depth distribution with a 1 km bin size, without taking magnitude completeness or cut-off magnitude into account. If the number of seismic events within 50 km of each grid exceeds 50, H is generated randomly based on its PDF, which is normalized from the frequency-depth distribution. Otherwise, H is taken as the average depth. When no earthquakes are reported around the grid, H is set to the default value of 5 km. Previous studies have shown that V_p (Sun and Toksöz 2006; Teng et al. 2013; Wei et al. 2016; Han et al. 2022) and V_s (Li et al. 2013; Bao et al. 2015; Shen et al. 2016; Tao et al. 2018; Xin et al. 2019; Han et al. 2022) of the crust within a depth of 60 km in Chinese mainland exhibit slightly larger values in the east than in the west, with a boundary at approximately 105°E . Therefore, we assigned V_p and V_s in the eastern region ($> 105^\circ\text{E}$) as random numbers following the uniform distributions $U(5.5, 7.5)$ and $U(3.7, 4.4)$, respectively, while in the western region ($\leq 105^\circ\text{E}$), the distributions are $U(4.5, 6.5)$ and $U(3.2, 3.9)$, respectively. In the five key EEW regions, earthquake focal depths are typically confined to within the first 50 km. The boundaries of these uniform distributions for V_p and V_s are derived from findings within a depth of 60 km presented by Han et al. (2022). T comprises the early data window (typically 3 s) after the P -wave onsite and the time for processing and transmitting data. For T , we generate random samples assuming a normal distribution $N(4, 0.5^2)$ for the new network and $N(5, 0.5^2)$ for the previous network, respectively. For information about the current system trigger criteria of the new network that determines T , we refer to the Discussion.

After performing Monte Carlo random sampling, the 50,000 samples of r on each grid exhibit a frequency distribution that resembles a normal distribution. As a result, we will use the mean and standard deviation of these samples to describe the spatial distribution of blind zones.

3.3 Map of blind zone radius

Figures 2a and 3a display the spatial distribution of the mean estimated r for the previous and new networks, respectively. In the previous network, areas with mean $r \leq 30$ km are mainly limited to major cities such as the Beijing-Tianjin-Hebei (Capital Region) and Guangdong-Hong Kong-Macau Greater Bay Area. However, the new network is expected to achieve almost complete coverage of areas with mean $r \leq 30$ km in North China, the southeast coast, the central China north-south seismic belt, and the middle section of the Tianshan Mountains in Xinjiang. This represents a significant improvement, with coverage increasing by a factor of 13 from 1.7% to reach 22%. The cumulative percentage distribution of mean estimated r in Fig. 3a indicates that about half of the region inside Chinese mainland will have a mean $r \leq 50$ km. In 15% and 38% of the area inside Chinese mainland, the mean r of the new network is expected to be smaller than 25 km and 40 km, respectively, which is of great significance for the successful functioning of the EEW system. Figure S4 illustrates the map of relative changes in the mean r inside Chinese mainland between the previous and new networks, as presented in Figs. 2a and 3a. Positive values denote a relative reduction in the mean r of the new network compared to the previous network. In some areas, negative values appear, representing a relative increase in the mean r for the new network compared to the previous network. This occurs because, although the total number of seismic stations has increased across Chinese mainland, changes in station locations—such as the removal or relocation of some stations—have led to an increase in the blind zone radius in those specific regions.

The reduction in the blind zone radius of the new network compared to the previous network can be used to calculate the theoretical increase in lead time ΔT , or available warning time, resulting from the improvement, based on Eq. (2). Assuming that the previous network has a blind zone radius r_{previous} and system alert trigger latency $T_{\text{tri}}^{\text{previous}}$, which is the time when the system first triggers an alert relative to the origin time, and that the new network has a blind zone radius r_{new} and system alert trigger latency $T_{\text{tri}}^{\text{new}}$, the additional lead time ΔT provided by the new network for an earthquake of the same depth H under identical site conditions (same V_s) can be calculated using the following equation:

$$\Delta T = T_{\text{tri}}^{\text{previous}} - T_{\text{tri}}^{\text{new}} = \frac{\sqrt{r_{\text{previous}}^2 + H^2} - \sqrt{r_{\text{new}}^2 + H^2}}{V_s} \quad (4)$$

Figure 4 illustrates the increase in lead time of the new network compared to the previous network. The results show that, compared to the previous network layout, lead times are expected to increase by approximately 20 s, 10 s, and 5 s for 50%, 30%, and 10% inside Chinese mainland, respectively. This significant improvement highlights the enhanced effectiveness of the new seismic network in providing earlier warnings.

Figures 2b and 3b display the spatial distribution of the standard deviation of estimated r for the previous and new networks, respectively. The maps show that there is a strong spatial

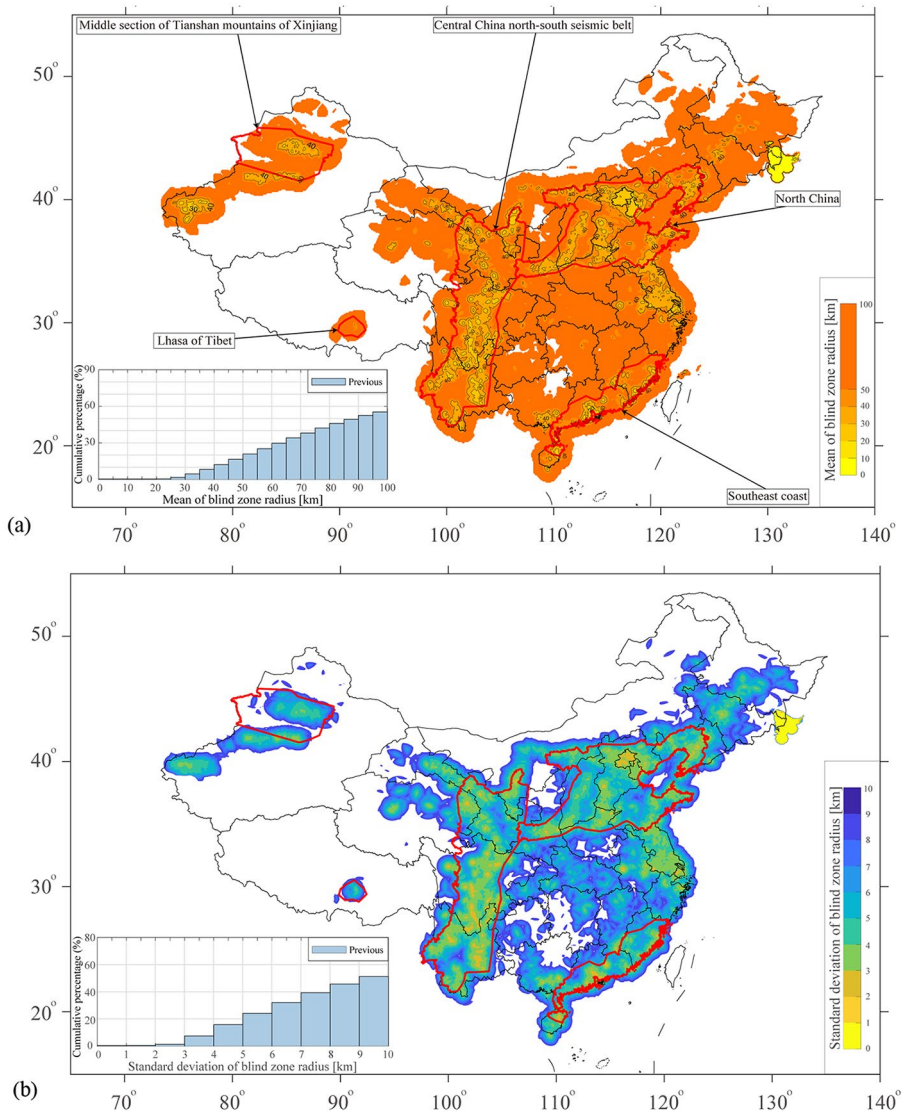


Fig. 2 Maps showing the (a) mean and (b) standard deviation of the estimated blind zone radii for the previous network, based on a minimum number of three trigger stations and a grid of $0.1^\circ \times 0.1^\circ$ resolution. Areas where the mean and standard deviation of blind zone radii surpass 100 km and 10 km, respectively, are displayed in white. The probability density function (PDF) of the blind zone radius has been constructed using Monte Carlo random sampling, with 50,000 iterations. Further details on the creation of the PDF is available in the text. The histograms in the bottom left depict the cumulative percentage distribution of mean and standard deviation of the blind zone radii inside Chinese mainland. The bins for the cumulative percentage distribution are 5 km for the mean and 1 km for the standard deviation, respectively. For this study, we define the “inside” of Chinese mainland as the region within 100 km outside of Chinese mainland’s boundaries (Figure S1). The areas delineated by the red solid lines represent the five key earthquake early warning (EEW) regions that were developed by the National System for Fast Report of Intensities and Earthquake Early Warning Project of China and are expected to be operational after 2024

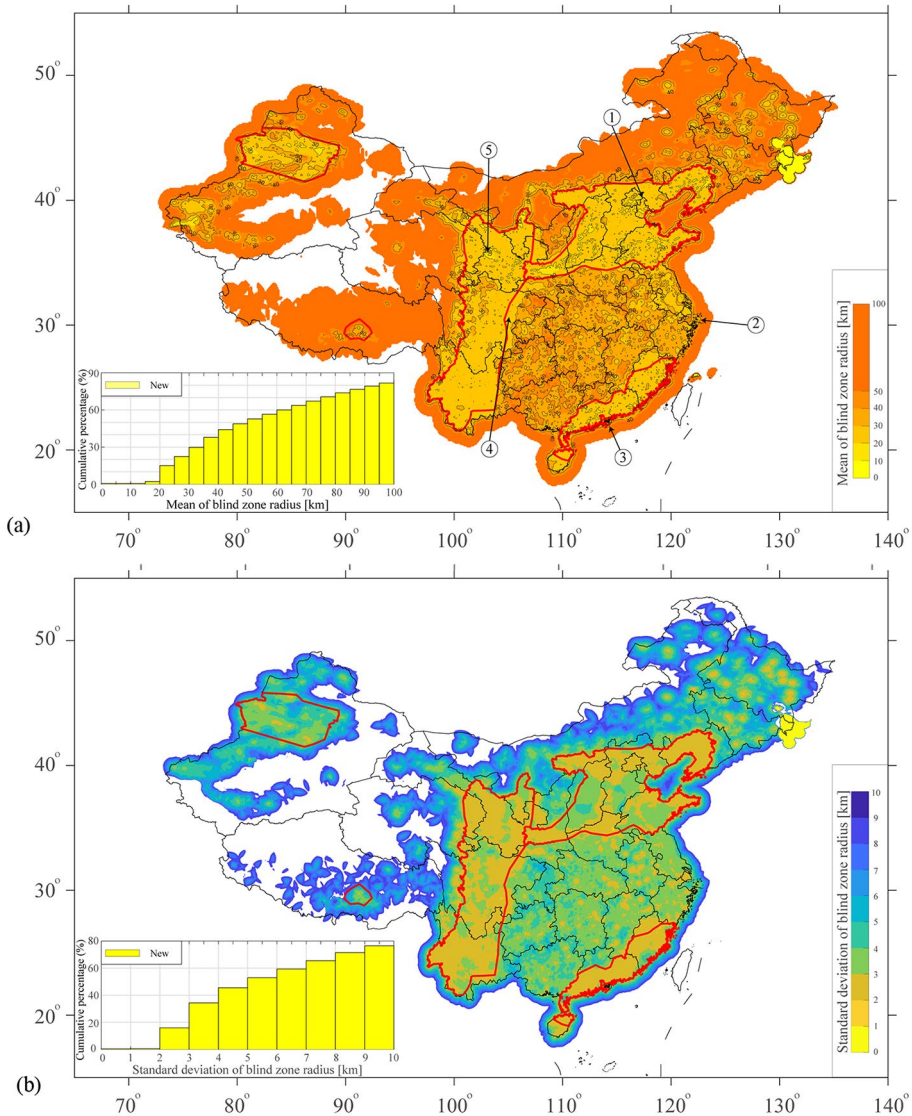


Fig. 3 Maps showing the (a) mean and (b) standard deviation of the estimated blind zone radii for the new network. The areas delineated by the red solid lines represent the five key EEW regions. Details are the same as Fig. 2. In (a), the labels 1 to 5 correspond to the following city clusters: (1) Beijing-Tianjin-Hebei (Capital Region), (2) Yangtze River Delta (Yangtze River Delta Urban Agglomeration), (3) Guangdong-Hong Kong-Macau Greater Bay Area, (4) Chengdu-Chongqing Urban Agglomeration, and (5) Lanzhou-Xining Urban Agglomeration

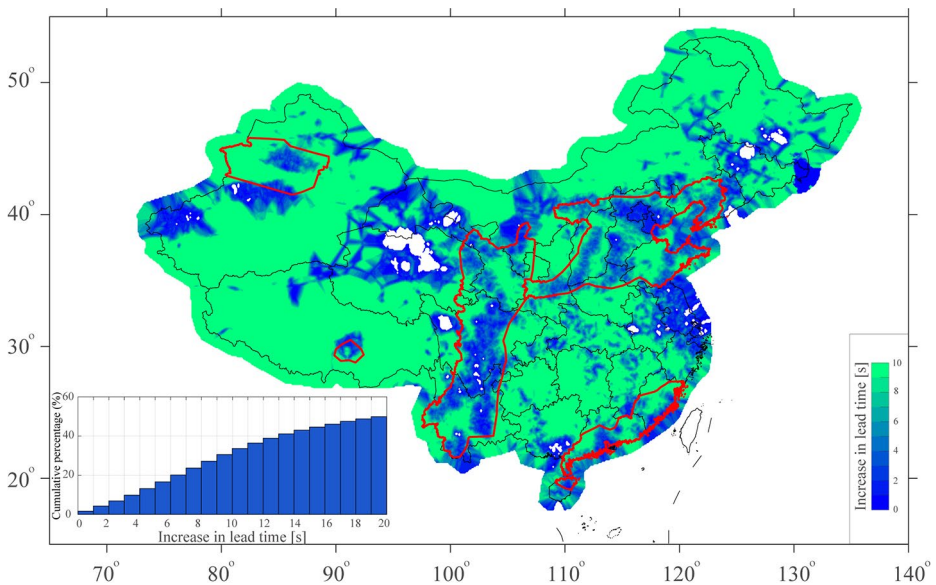


Fig. 4 Map showing the increase in lead time (available warning time) of the new network compared to the previous network, assuming a minimum of three trigger stations. The areas delineated by the red solid lines represent the five key EEW regions. The histograms in the bottom left illustrate the cumulative percentage distribution of the lead time increase within Chinese mainland. Areas where the lead time is reduced are displayed in white

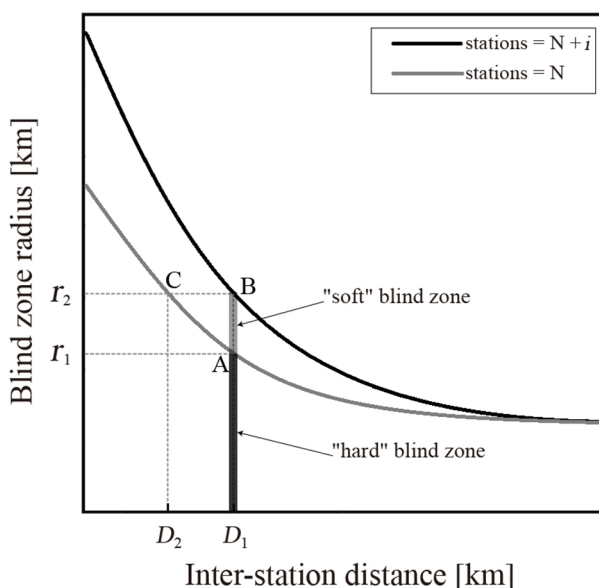
correlation between the standard deviation of r with station density. Overall, areas where the standard deviation of r is less than 5 km and 10 km will cover 46% and 77% of the areas inside Chinese mainland for the new network, respectively. For the previous network, these proportions are 16% and 51%, respectively. This implies that densification of the networks will effectively constrain the potential variability in the size of the blind zones.

It should be noted that, for both the previous and new networks, there are large areas with a mean and standard deviation of zero in Northeast China (bordering Russia and North Korea). This is mainly due to the fact that the focal depth of earthquakes observed in this region is generally greater than 400 km. The Western Pacific Plate subducts beneath the Eurasian Plate and penetrates about 600 km deep into the mainland of Northeast China through the Japan Trench, forming the only deep-focal seismic belt in China (Zhang and Tang 1983; Gudmundsson and Sambrige 1998; Sun and He 2004).

4 “Soft” blind zone and its control

Kuyuk and Allen (2013) used a series of evenly distributed modeled shallow point-sources at a depth of 8 km and stations to determine a semi-quantitative empirical curve that describes the relationship between the minimum number of trigger stations, the inter-station distance D and r . Figure 5 ($D_1 < D_2$; $r_1 < r_2$) shows that r decreases dramatically with the decrease of inter-station distances when the network density of seismic stations is small. However, for a very dense network, r converges to a plateau, which is approximately the same for differ-

Fig. 5 The relationship between the blind zone radius (r) and inter-station distance (D) for various numbers of trigger stations. The x -axis represents decreasing inter-station distances from left to right ($D_1 < D_2$) corresponding to an increasing network density, and the y -axis increases from bottom to top ($r_1 < r_2$). Modified from Kuyuk and Allen (2013), Fig. 5, with simulations assuming earthquakes at a depth of 8 km



ent minimum number of trigger stations, ranging from 10 to 20 km depending on different conditions (Kuyuk and Allen 2013).

Figure 5 can also be used to illustrate the equivalence principle between the minimum number of trigger stations and network density from the perspective of blind zones. Assuming a seismic network with an inter-station distance of D_1 , the minimum number of N trigger stations corresponds to a blind zone of radius r_1 , as shown in point A of Fig. 5. This ideal blind zone with radius r_1 is called the theoretical “hard” blind zone by Li et al. (2016). However, in the real world, some i stations near the epicenter may fail to work properly (Bormann 2002) or may not be triggered due to an underestimation of ground-shaking (e.g., YZP station during the 2008 M_s 8.0 Wenchuan earthquake; Ma et al. 2012). Therefore, the EEW system has to adopt i more stations to be triggered, which corresponds to point B with a blind zone radius r_2 (where $r_1 < r_2$) in Fig. 5. This unexpected and larger blind zone with radius r_2 is referred to as the “soft” blind zone by Li et al. (2016). From Fig. 5, the size of this “soft” blind zone is equivalent to the size of the “hard” blind zone of a seismic network adopting a minimum number N of trigger stations with inter-station distance of D_2 (where $D_1 < D_2$), as shown in point C.

To investigate the performance of the new EEW system of China (assuming $N=3$, same as Fig. 3) affected by the “soft” blind zone, we show the cumulative distribution of the mean and standard deviation of estimated r inside Chinese mainland for $i=1$ (corresponding to a minimum number of four trigger stations) and $i=2$ (corresponding to a minimum number of five trigger stations) potential station failures, respectively (Fig. 6). The map showing the mean and standard deviation of r for $N+i=4$ and 5 can be found in Figure S5. The areas with mean $r \leq 20$ km, 30 km, and 40 km account for 1%, 19%, and 32% of the total area inside Chinese mainland for $N+i=4$, and for 0.6%, 17%, and 27% for $N+i=5$, respectively. Compared with Fig. 3a (without station failures), the area with mean $r \leq 30$ km is reduced by 15% and 25% for $N+i=4$ and 5, respectively. The proportion of areas with mean $r \leq 20$ km and 40 km decreases by 55% and 16% for $N+i=4$, and 71% and 28% for $N+i=5$, respec-

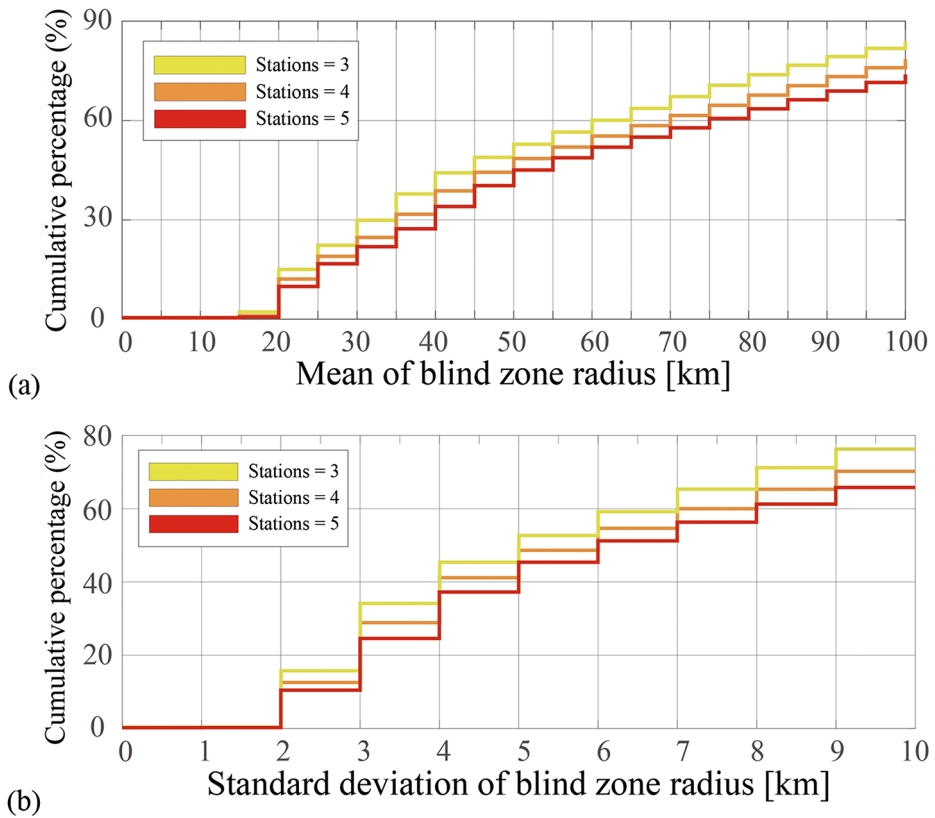


Fig. 6 The cumulative percentage distribution of the (a) mean and (b) standard deviation of the blind zone radii inside Chinese mainland for the cases of three, four, and five trigger stations, respectively. The bins for the cumulative percentage distribution are 5 km for the mean and 1 km for the standard deviation, respectively

tively (Fig. 6a). Compared with Fig. 3b, the area with the standard deviation of r smaller than 5 km is reduced by 9% and 18% for $N+i=4$ (from 45.5 to 41.25%) and 5 (from 45.5 to 37.47%), respectively (Fig. 6b). Overall, continuing to densify the seismic network in some regions where r ranges from 15 to 40 km is still recommended to control the “soft” blind zones and minimize blind zones.

5 Earthquake early warning effectiveness in five key eew regions

The CEA aims to establish five key EEW regions through the implementation of the National System for Fast Report of Intensities and Earthquake Early Warning Project. The five regions shown in Fig. 2a include North China (with an area of 712,747 km², a population of 316 million, accounting for 23.6% of the national population, and a GDP [Gross Domestic Product] of 13.05 trillion yuan, accounting for 20.51% of the national GDP), the central China north-south seismic belt (850,905 km², a population of 109 million, accounting for 8.2% of the national population, and a GDP of 5.15 trillion yuan, accounting for

8.09%), the southeast coast of China (236,502 km², a population of 108 million, accounting for 8.07% of the national population, and a GDP of 8.12 trillion yuan, accounting for 12.76%), the middle section of the Tianshan Mountains in Xinjiang (235,979 km², a population of 9 million, accounting for 0.68% of the national population, and a GDP of 0.47 trillion yuan, accounting for 0.74%), and Lhasa in Tibet (30,438 km²). Collectively, these regions account for approximately 40% of the national population and the national GDP, and have experienced earthquakes with $M \geq 6$ in the past (Figure S1). These data were collected up to 2016 and are from the Feasibility Report of the National System for Fast Report of Intensities and Earthquake Early Warning Project of China, which was released within the CEA (courtesy of Changsheng Jiang for providing this report). The report does not include GDP and population data for Lhasa in Tibet, and the present paper does not have the capacity to add such data or update all figures to more recent values (e.g., 2024). This limitation arises because the key EEW regions do not correspond neatly to specific provinces, cities, or counties but span multiple administrative boundaries. Without fine-grained data (e.g., at the county or township level), conducting further statistical work remains challenging. Over the past decade, Chinese mainland's total population has seen only a small increase by 3.5% (from 1.369 billion in 2013 to 1.41 billion in 2023; see Data and Resources), while GDP has doubled (approximately from 56.88 trillion in 2013 to 126.06 trillion yuan in 2023; see Data and Resources). Despite these changes, the percentages of the national population and GDP represented by the key EEW regions have likely remained relatively stable, making the data in the 2016 report a valuable reference. According to the fifth-generation hazard zoning map in China (see Data and Resources), the projected peak ground accelerations (PGA) with an exceedance probability of 10% in the next 50 years are expected to reach 0.1 g across almost the entire area of the five regions, with some parts even reaching 0.3 g.

Figures 7 and 8 display the cumulative distribution of the mean and standard deviation of the estimated r for the new network in the five regions, as spatially presented in Fig. 2. We find that the areas with mean $r \leq 30$ km in North China and the southeast coast account for approximately 98% of the regions. Areas with a mean r smaller than 20 km even account for 14% of the regions. The standard deviation of r in the two regions is less than 5 km, indicating a small variability of r due to the high density of the network. The central China north-south seismic belt ranks third, with mean $r \leq 30$ km and 20 km accounting for 91% and 5%, respectively, and a standard deviation smaller than 5 km across the entire region. However, the middle section of the Tianshan Mountains in Xinjiang and Lhasa in Tibet exhibit slightly weaker EEW performance. In the middle section of the Tianshan Mountains in Xinjiang, the coverage of mean $r \leq 30$ km and 20 km is 63% and 6%, respectively, with a standard deviation below 5 km in 87% of the region. Lhasa in Tibet has a relatively poor EEW ability with only 3% coverage of mean $r \leq 50$ km, and the coverage of the 5 km and 10 km standard deviations is 2% and 87%, respectively, making it difficult to achieve effective EEW. These represent the theoretical “hard” blind zone for the new network, assuming a minimum of three trigger stations.

In addition, Figs. 7 and 8 illustrate the impact of the “soft” blind zone on EEW performance in five regions for $i=1$ and $i=2$ failures of stations closest to the epicenter. In North China and the southeast coast, the areas with mean $r \leq 30$ km are reduced by 5% and 2%, and areas with mean $r \leq 20$ km decreased by 90% and 70% when the number of station failures increases from $i=1$ to $i=2$, compared to the case of $i=0$ ($N+i=3$). In the central China north-south seismic belt, the reductions are approximately 10–20% for mean $r \leq 30$ km and

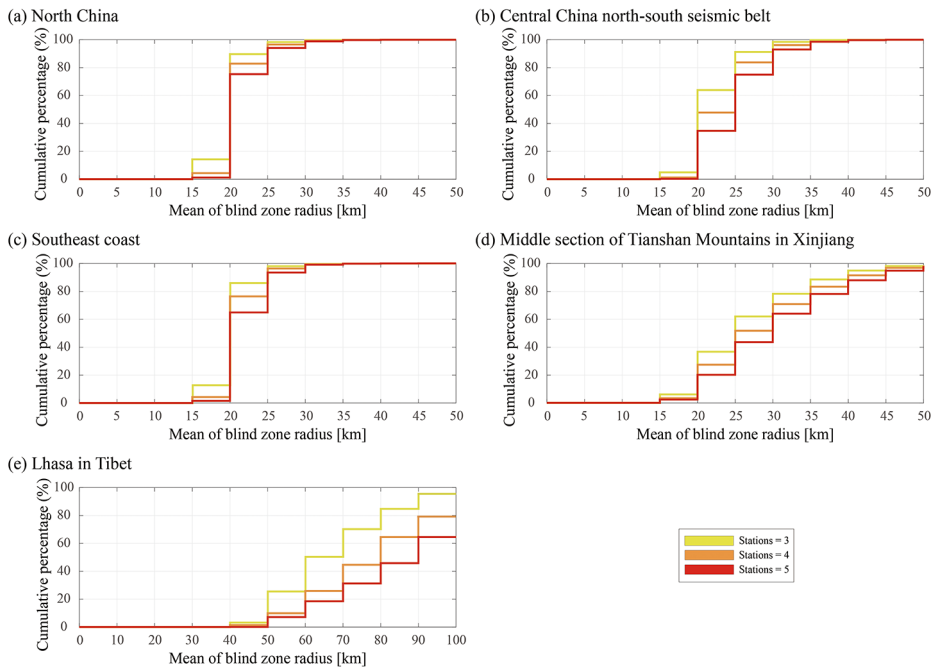


Fig. 7 Cumulative percentage distribution of mean blind zone radii in five key EEW regions for the cases of three, four, and five trigger stations, respectively. The bins for the cumulative percentage distribution are 5 km

80–90% for mean $r \leq 20$ km. The middle section of the Tianshan Mountains in Xinjiang experiences greater reductions, ranging from 20 to 30% for mean r smaller than 30 km and 50–60% for mean $r \leq 20$ km. In Lhasa, there is no area with mean $r \leq 50$ km in the presence of two station failures. In North China, the southeast coast and the central China north-south seismic belt, the areas with a standard deviation of r smaller than 5 km do not change significantly when station failures occur. Therefore, it is crucial to continuously densify the network to control the “soft” blind zone for optimal functioning of the EEW system in the five regions.

6 Discussion

Globally acknowledged for their dense seismic monitoring networks and efficient EEW systems, regions like Japan, the U.S. West Coast, and Taiwan have set comparable standards. These regions have achieved approximately 30 km of comprehensive coverage, both for inter-station distances and blind zones (e.g., Kuyuk and Allen 2013; Kohler et al. 2020). Similarly, the implementation of an advanced network in key EEW regions spanning North China, the central China north-south seismic belt, the southeast coast, and the middle section of the Tianshan Mountains in Xinjiang will result in a network density and coverage area with small-radius blind zones that can rival even the most extensive seismic networks worldwide, such as those in Japan, the U.S. west coast, and Taiwan. This enhanced network

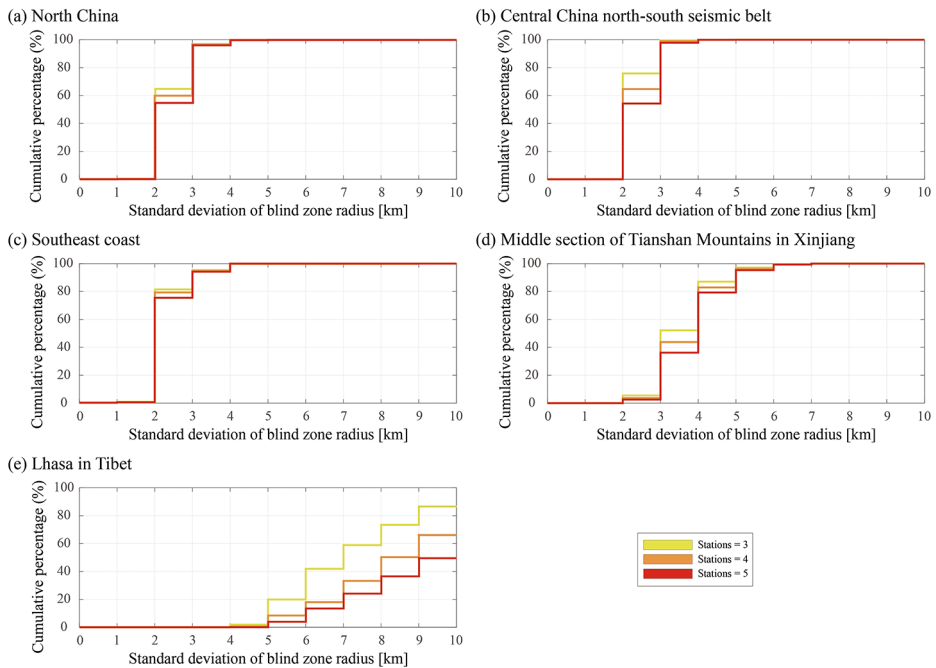


Fig. 8 Cumulative percentage distribution of standard deviation of the blind zone radii in five key EEW regions. The bins for the cumulative percentage distribution are 1 km

will offer a 30 km inter-station distance and blind zones in all key EEW regions, with the potential to cover key urban areas within a range of 10 to 20 km. Compared to the previous network, in the four key EEW regions excluding Lhasa in Tibet, the coverage area with a 30 km inter-station distance increases from approximately 33–96%, nearly a threefold increase. The coverage percentages for 20 km and 10 km inter-station distances increase from 14 to 88% (a 6.5-fold increase) and from 2 to 52% (a 30-fold increase), respectively. In Lhasa of Tibet, the coverage area with a 40 km inter-station distance remains almost unchanged, while the coverage area with a 50 km inter-station distance increases from the current 0.4–27%.

In this section, we will first conduct a preliminary cost-outcome evaluation of the new EEW system from the perspective of blind zones. Then, using the empirical regression relationships previously established to estimate the system alert trigger latency T_{tri} based on inter-station distances, we will predict the T_{tri} of the new network. This serves as an independent prediction result, which we will compare with our earlier blind zone radius predictions.

6.1 The cost-outcome evaluation

The costs per station of 1,000 new datum, 2,100 new basic, and 10,000 new ordinary stations of the new network are 235,000 RMB, 130,000 RMB, and 10,000 RMB, respectively (Table 1). Compared to the cost of building previous broadband seismic (350,000 RMB) and strong motion (200,000 RMB) stations in 2008, the station construction cost has decreased.

Table 1 Costs of the new stations in the new seismic network in Chinese mainland

Items	DT station	BS station	OD station
The site construction	90,000	60,000	N/A
The data acquisition system	50,000	40,000	N/A
A broadband seismometer	60,000	N/A	N/A
An accelerometer	15,000	15,000	N/A
A MEMS sensor	N/A	N/A	10,000
other accessories	20,000	15,000	N/A
Total	235,000	130,000	10,000

DT station: datum station;
BS station: basic station; OD
station: ordinary station. Costs
are YUAN in RMB. Data from
Qiang Ma and Changsheng
Jiang (written comm. 2021)

This is mainly due to the localized development and production of seismometers, as well as the decline in their prices. The upgrade costs per station (to replace the data acquisition system and seismometer) for 1,000 previous broadband seismic and 1,100 strong motion stations are 125,000 RMB and 55,000 RMB, respectively. These stations correspond to the high-gain broadband seismic and strong motion stations of the previous network shown in Figure S2a. The majority of these stations were upgraded and incorporated into the new network. Therefore, the cost directly used to increase station density is 794 million RMB, which accounts for approximately 42% of the total investment (1.87 billion RMB).

Figures 2 and 3 show that the area covered by the theoretical “hard” blind zones smaller than 30 km will increase from 1.7 to 22.3% inside Chinese mainland (11.5 million km²), which means that this area will increase by 2.4 million km² (almost the size of Algeria). Assuming the improvement is made through the densification of the seismic network with an investment of 794 million RMB, we estimate that every 1,000,000 RMB (146,000 USD in 2023) invested in the new networks will increase the area where the blind zone radius is smaller than 30 km by 3,000 km². Similarly, every 1,000,000 RMB invested will increase the areas where the blind zone radius is smaller than 20 km and 40 km by 250 km² and 4,300 km², respectively. The cost-outcome evaluation outlined above is admittedly rudimentary, as the computation hinges on a substantial assumption: a linear correlation between investment and the expansion of the area covered by blind zones of small radii. The credibility of this assumption naturally hinges on the specific placement of new seismic stations.

6.2 The system alert trigger latency

In this study, the theoretical prediction of r for the new network is based on the decomposition of the system alert trigger latency T_{tri} in terms of t_p and T . t_p is determined by Δ , H and V_p , while T is hypothesized to follow a normal distribution. Li et al. (2021a) proposed the following empirical relationship for a line-source model-based EEW algorithm (Finite-Fault Rupture Detector algorithm, FinDer; Böse et al. 2012, 2015, 2018) in the Sichuan-Yunnan region, China:

$$T_{\text{tri}} = 2.5 \exp(0.04D_{\text{epi}}) \quad (5)$$

where T_{tri} is counted from the origin time of the earthquake with an inter-station distance D_{epi} at its epicenter. The relationship is based on a playback investigation of 58 earthquakes ($5.0 \leq M \leq 8.0$). The algorithm uses a trigger threshold of 4.6 cm/s² and assumes a minimum number of three trigger stations. Additionally, it is assumed that there is no data transmission latency.

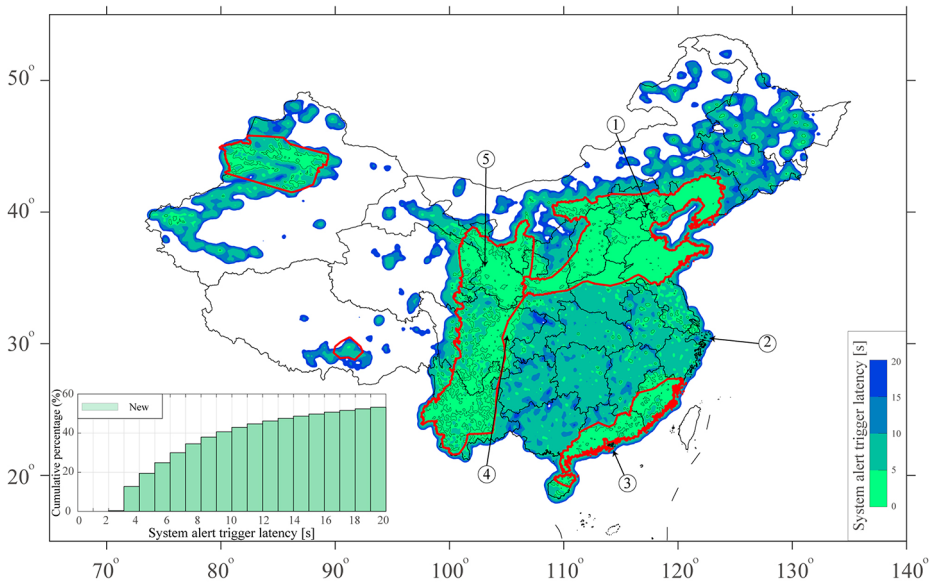


Fig. 9 Map showing the estimated system alert trigger latency T_{tri} from the earthquake origin time for the new EEW system. Areas where T_{tri} surpasses 20 s are displayed in white. The T_{tri} was calculated using Eq. (5). The areas delineated by the red solid lines represent the five key EEW regions. The black thin solid line represents the contour for $T_{tri} = 4$ s. The labels 1 to 5 correspond to the following city clusters: (1) Beijing-Tianjin-Hebei (Capital Region), (2) Yangtze River Delta (Yangtze River Delta Urban Agglomeration), (3) Guangdong-Hong Kong-Macau Greater Bay Area, (4) Chengdu-Chongqing Urban Agglomeration, and (5) Lanzhou-Xining Urban Agglomeration. The histogram in the bottom left shows the cumulative percentage distribution of T_{tri} inside Chinese mainland

Equation (5) provides an alternative method for estimating T_{tri} . Previous studies have shown that T_{tri} values obtained with other point-source-based algorithms differ slightly from those obtained with the FinDer algorithm (e.g., Li et al. 2020; Kohler et al. 2020). Therefore, we propose that Eq. (5) can be adopted for EEW systems with the same trigger configurations to estimate the system alert trigger latency with the same uncertainty of $\sigma_{T_{tri}} = 1.8$ s as determined by the regression conducted by Li et al. (2021a). Figure 9 shows that the areas with empirical T_{tri} smaller than 5 s and 10 s account for 20% and 41%, respectively, which is close to the theoretical results for mean r smaller than 30 km (22%) and 45 km (44%) in Fig. 3a.

Other research groups also reported the performance of the forthcoming EEW system in Chinese mainland that aligns well with our predictions. Pan et al. (2019) anticipated a system alert trigger latency of 7 to 12 s in the southeastern region of Gansu, China (approximately at the location labeled “5” in Fig. 9), equating to a blind zone radius of 20 to 40 km (considering $V_s = 3.57$ km/s and $H = 15$ km in their study). Our results in Fig. 3a indicate a blind zone radius spanning 20 to 30 km for the same region, with an estimated uncertainty of 4 km. Peng et al. (2021) analyzed the EEW performance of the new network during the earthquake sequence in Changning, Sichuan, between 2019 and 2020 (slightly below the location labeled “4” in Fig. 9). They reported that the system alert trigger latency T_{tri} was 6.3 ± 3.5 s (equivalent to a blind zone radius of 21 ± 13 km, assuming $V_s = 3.5$ km/s and $H = 7$ km). This reporting aligns reasonably well with our predicted values of T_{tri} (4 ± 1.8 s)

in Fig. 9 and the blind zone radii r (21 ± 3 km) in Fig. 3a for the same region. However, there was a slightly larger deviation in the results reported by Peng et al. (2021), likely stemming from the complexities of the operational system that extend beyond the simplifications we made in our parameter assumptions.

During the trial operations of the new EEW system, observed values of T_{tri} across various regions align well with our predictions. For instance, in Xinjiang, approximately 50% of the region is reported to have a T_{tri} of less than 10 s (corresponding to a blind zone radius of approximately 35 km; Yan et al. 2023). For three notable earthquakes in this region—the 2021 Baicheng M_S 5.4, 2022 Changji M_S 4.8, and 2024 Wushi M_S 7.1 earthquakes—the reported T_{tri} values were approximately 4 s, 8 s, and 12 s (corresponding to blind zone radii of approximately 15 km, 30 km, and 40 km), respectively. Our predictions, as shown in Fig. 9 for T_{tri} and Fig. 3a for blind zone radius r , are in close agreement with these observations. In the Sichuan and Yunnan regions, reports show that T_{tri} typically ranges from 5 to 8 s (corresponding to blind zone radii of approximately 20 to 30 km; Li et al. 2024b; Guo et al. 2024). For the 2021 Qinghai Maduo M_S 7.4 earthquake, which occurred 55 km outside the Sichuan seismic network, the system successfully issued a regional warning with a T_{tri} of 12 s (corresponding to a blind zone radius of approximately 40 km; Li et al. 2021b). These results align well with our predicted T_{tri} values and blind zone radii for similar regional warnings. In Hebei and Shanghai, observed T_{tri} values range from 4 to 10 s (corresponding to blind zone radii of approximately 15 to 35 km; Wang et al. 2023; Shao et al. 2024), consistent with our model's estimates for onsite warnings (Figs. 3a and 9). Overall, during trial operations in the key EEW regions, T_{tri} ranged from 4 to 10 s for onsite warnings (corresponding to blind zone radii of approximately 15 to 35 km) and slightly exceeded 10 s for regional warnings, depending on the epicentral distance from the network edge. These observed values provide strong validation for our predicted results, underscoring the robustness of the new network's configuration and the reliability of our modeled blind zone radii r and T_{tri} .

The current test of the new EEW system in the Chinese mainland implements a relatively complex set of system alert trigger criteria to mitigate false alerts. Firstly, different trigger thresholds have been set for each type of station based on their data observation quality, starting with datum stations having the best quality, followed by basic stations, and lastly ordinary stations with the worst quality. Secondly, the system applies a double-checking mechanism to ensure that the alert triggers are not caused by non-earthquake events. This is done by analyzing the signal-to-noise ratio of the data, the waveform duration of events, and the spatial configuration of trigger and non-trigger stations in real-time. Finally, a system alert is issued only after confirming that an earthquake occurred. Despite the complexity of this procedure, the system latency T is reasonably small based on practical test runs. Therefore, T is modeled to follow a normal distribution $N(4, 0.5^2)$ for the new network in this study.

However, it is worth noting that the system alert trigger criteria are not directly used or considered in our calculation. Rather, their significance is indirectly reflected in the simulation of T . If the current alert trigger criteria remain unchanged, our simulation for T , and therefore the estimation of r , is reasonable. On the one hand, our estimation of the blind zone size based on the simulation of T and a minimum number of three trigger stations is conservative. Typically, the system would trigger an alert once two stations are triggered, and source parameters can be estimated using the early part of the event waveform in the

test run of the new network. On the other hand, our estimation of the blind zone based only on T may be optimistic and correspond to a lower bound. Indeed, the size of the blind zone could be larger than our theoretical calculation if additional latency is considered, such as communication and response latency needed to take efficient actions. It is important to note that the blind zone evaluated in our study is determined from the properties of the EEW system rather than from considerations involving earthquake risk, which should consider additional factors related to the earthquake impact on society. Whatever the end use, the blind zones determined from the EEW system is useful in guiding the optimization of system construction and earthquake risk reduction strategies.

7 Conclusions

By simply densifying the seismic network layout, the performance of an EEW system can be significantly improved. As of 2024, approximately 80% of the area inside Chinese mainland will have inter-station distances of less than 100 km, with 50%, 30%, and 17% having inter-station distances of 50 km, 25 km, and 15 km, respectively. These proportion of inter-station distances are expected to expand by factors of 1.4, 2.2, 4.4, and 10 in the near future, compared to that of the previous layout. In this study, we predicted the spatial distribution of the blind zone radii and quantified the warning effectiveness of the new EEW system in Chinese mainland based on its network configuration. As a novelty, we employed Monte Carlo random sampling with 50,000 iterations to account for uncertainty in the estimate of r . Additionally, we investigated the impact of “soft” blind zones caused by potential station failures on the performance of the new EEW system in China.

The results demonstrate that r is inversely correlated only with H and V_p , and H contributes significantly to the occurrence of zero blind zones. Taking into account the practical change amplitude scale of parameters related to r in operational system, Δ and H have the greatest impact on r , resulting in a magnitude change of r of 10^1 km and -10^1 km. The area covered by blind zones smaller than 30 km will increase by a factor of 13 from 1.7 to 22% inside Chinese mainland, which means that this area will increase by 2.4 million km² (almost the size of Algeria). This improvement is a positive outcome of the new system, as it reflects a substantial reduction of the largest blind zones. Compared to the previous network layout, lead times are expected to increase by approximately 20 s, 10 s, and 5 s for approximately 50%, 30%, and 10% inside Chinese mainland, respectively. The new network is expected to achieve almost complete coverage of areas with mean $r \leq 30$ km in North China (98%), the southeast coast (98%), and the central China north-south seismic belt (91%). However, the middle section of the Tianshan Mountains in Xinjiang (63%) and Lhasa in Tibet exhibit weaker EEW performance, and achieving effective EEW for Lhasa in Tibet is difficult. Additionally, areas with a standard deviation of r less than 5 km and 10 km will cover 46% and 77% of the areas inside Chinese mainland, respectively. This implies that the densification of the networks will effectively constrain the potential variability in the size of the blind zones, which is significant for the successful functioning of the nationwide EEW system. Considering that stations located near the epicenter may fail to function correctly or underestimate the hazards of ground-shaking, the proportion of areas with mean $r \leq 30$ km and 40 km can decrease by around 15% for one station failure and by around 25% for two station failures. To prevent the unforeseen enlargement of blind zones due to com-

mon station failures, it is still necessary to continue to densify the seismic network in some key regions where the blind zone radii range from 15 to 40 km.

The blind zone radius is one of the most effective metrics for evaluating the overall performance of an EEW system, but it comes with certain limitations that deserve consideration: (1) Algorithm dependency: The blind zone radius is inherently tied to the specific EEW algorithm used, particularly through the T parameter and the minimum number of triggered stations required. The parameterization of its calculation is highly context-dependent and influenced by the algorithm adopted, which can limit the generalizability of the findings to other regions or systems. (2) Simplified assumptions in equations: Eqs. (1) and (2) provide a simplified representation of the complex processes underlying EEW systems. For instance, wave velocities vary spatially rather than being uniform, and the system alert trigger latency may depend on the type and configuration of seismic sensors. Although we address these issues by incorporating uncertainties into the parameter values, the limitations of these simplifications should be acknowledged as they may affect the applicability of the equations. (3) Parameter selection: The values used for parameters in Eqs. (1) and (2) are pivotal to the study's conclusions. While these values are chosen empirically and represent typical conditions, it is important to justify these choices thoroughly to ensure the robustness and credibility of the results.

Our work not only offers insights into the anticipated performance of the forthcoming EEW network in Chinese mainland but also introduces a broadly applicable evaluation approach for predicting the performance of any EEW system during its planning, design, and implementation stages. This investigation serves as a valuable reference for the further enhancement and optimization of the EEW system in Chinese mainland. Further quantitative exploration of how expanding coverage area with blind zones of small radii (e.g., 30 km) could effectively mitigate earthquake risks by reducing exposure to population, the economy, essential lifelines, and infrastructure warrants consideration.

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Data availability The earthquake catalog and information for the previous and the new seismic networks were provided by the China Earthquake Networks Center (<https://doi.org/10.11998/SeisDmc/SN>) through the internal link of the Earthquake Cataloging System at China Earthquake Administration (<http://10.5.1>

60.18/console/index.action). The total investment information for the National System for Fast Report of Intensities and Earthquake Early Warning Project of China was obtained from the China Earthquake Networks Center (<https://www.cenc.ac.cn/cenc/zt/361404/361414/361563/index.html>, last accessed in January 2022). The latest updates on the progress of this project are available at CCTV News (https://content-static.cctvnews.cctv.com/snow-book/index.html?item_id=11557423601650128874%26toc_style_id=feeds_default%26share_to=wechat%26track_id=3a0ddb93-e1e5-4469-8e88-8857559fcab3, last accessed in September 2024). The 2023 population and GDP statistics for mainland China were sourced from the National Bureau of Statistics of China (https://www.stats.gov.cn/sj/zxfb/202402/t0240228_1947915.html, last accessed in September 2024). The fifth-generation earthquake hazard zoning map of China can be accessed through the link: <https://www.gb18306.net/> (last accessed in March 2023).

Declarations

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